
UNIVERSITY PROJECT · SUPPLY CHAIN ANALYTICS

Logistics Performance, Revenue & Delivery Analysis

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Supply Chain Analytics Project · SQL & Power BI

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Abstract

Supply chain management plays a critical role in modern organizations by ensuring that products are delivered efficiently and on time. This project analyses logistics operations, delivery performance, profitability, and customer distribution using the **DataCo Supply Chain Dataset**.

MySQL was used for data cleaning, KPI creation, and business analysis, while **Power BI** was used to build interactive dashboards for visualisation and decision support.

The project focuses on identifying delivery inefficiencies, analysing revenue and profit generation, evaluating late-delivery risk, and providing actionable recommendations to improve operational performance.

Keywords: Supply Chain Analytics, Logistics Performance, Delivery Efficiency, Power BI Dashboard, MySQL, Late Delivery Risk, KPI Development.



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1.1 Background

Supply chains connect suppliers, warehouses, transportation systems, and customers across global markets. Efficient supply chain operations are essential for reducing costs, improving customer satisfaction, and maintaining competitive advantage in today's demanding retail environment.

Data analytics allows organisations to monitor performance continuously, identify inefficiencies before they compound, and make informed, evidence-based business decisions. The combination of relational database querying and interactive dashboard development provides a powerful analytical toolkit for supply chain professionals.

1.2 Project Objectives

- Obj. 1.** Analyse logistics and delivery performance across product categories and geographies.
- Obj. 2.** Evaluate total revenue and profitability at the organisational and category level.
- Obj. 3.** Measure shipping delays by computing the `Delay_Days` metric.
- Obj. 4.** Assess late-delivery risk and identify high-risk categories.
- Obj. 5.** Build a professional, interactive four-page Power BI dashboard.
- Obj. 6.** Generate actionable business recommendations grounded in analytical findings.

1.3 Tools and Technologies

Table 1.1. Tools Used in This Project

Tool	Purpose	Version
MySQL	Data storage, cleaning, KPI queries	8.0
Power BI Desktop	Interactive dashboard development	Latest
DataCo Dataset	Source transactional data	Kaggle v5
L ^A T _E X	Report typesetting	pdfLaTeX

Dataset Description

2.1 Dataset Overview

The **DataCo Global Supply Chain Dataset** is a large-scale, publicly available transactional dataset representing real-world supply chain operations for a multinational retail company. It contains over **180,000 records** and **53 variables** covering customer orders, product categories, shipping operations, delivery performance, and financial metrics.

Table 2.1. Dataset Summary Statistics

Attribute	Value
Total Records	180,519 rows
Number of Columns	53 original features
Geographic Coverage	40 + countries, 5 market regions
Product Categories	24 categories across 5 departments
File Format	CSV (imported to MySQL)
Dataset Source	Kaggle — DataCo Smart Supply Chain

2.2 Dataset Preview

```

15 • SELECT *
16 FROM supply_chain
17 LIMIT 10;
18 • SHOW TABLES;
19 • DESCRIBE supply_chain;
20

```

Type	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Delivery Status	Late_delivery_risk	Category Id	Category Name
DEBIT	3	4	91.25	314.6400146	Advance shipping	0	73	Sporting
TRANSFER	5	4	-249.0899963	311.3599854	Late delivery	1	73	Sporting
CASH	4	4	-247.7799988	309.7200012	Shipping on time	0	73	Sporting
DEBIT	3	4	22.86000061	304.8099976	Advance shipping	0	73	Sporting
PAYMENT	2	4	134.2100067	298.25	Advance shipping	0	73	Sporting
TRANSFER	6	4	18.57999992	294.980011	Shipping canceled	0	73	Sporting
DEBIT	2	1	95.18000031	288.4200134	Late delivery	1	73	Sporting

Figure 2.1. Dataset Preview — First rows of the DataCo supply chain table

2.3 Key Variables

Table 2.2. Key Variables Used in the Analysis

Variable	Type	Description
Sales per customer	Float	Revenue generated per customer transaction
Benefit per order	Float	Profit generated per individual order
Days for shipping (real)	Integer	Actual number of days taken to ship
Days for shipment (scheduled)	Integer	Originally planned shipping duration
Delivery Status	Varchar	Categorical outcome of the shipment
Late_delivery_risk	Binary	Flag: 1 = high delivery risk
Category Name	Varchar	Product category label
Customer City	Varchar	Customer geographical city
Customer Country	Varchar	Customer geographical country
Department Name	Varchar	Department owning the product

2.4 Delivery Status Categories

Table 2.3. Delivery Status Distribution

Status	Count	Percentage	Flag
Late Shipping	98,975	54.8%	Critical
Advance Shipping	47,908	26.5%	Good
Shipping on Time	19,148	10.6%	Target
Shipping Cancelled	14,488	8.1%	Review
Total	180,519	100.0%	

**Risk Alert**

Only **10.6%** of orders arrive on time. The combined late and cancelled rate is **62.9%** of all shipments — a severe systemic logistics failure requiring immediate operational attention.

3.1 Analytical Framework

The project follows a structured four-phase analytical methodology:

- Phase 1. Data Acquisition & Preparation** — Import dataset into MySQL, validate data types, check for missing values and duplicates.
- Phase 2. Feature Engineering** — Create the derived `Delay_Days` metric and supporting categorical columns.
- Phase 3. SQL Exploratory Analysis** — Query-driven extraction of KPIs across financial, operational, geographic, and risk dimensions.
- Phase 4. Dashboard Development & Reporting** — Build four-page interactive Power BI dashboard; synthesise findings and recommendations.

3.2 Data Preparation

The dataset was imported into MySQL 8.0 and subjected to the following quality checks:

- ✓ **Missing values** — Null check on all key columns
- ✓ **Duplicate records** — Order ID deduplication check
- ✓ **Data type validation** — Numeric fields confirmed as INT/FLOAT
- ✓ **Data consistency** — Delivery status values normalised

3.2.1 Database Setup SQL

```
1 CREATE DATABASE SupplyChainAnalytics;  
2 USE SupplyChainAnalytics;  
3  
4 -- Record count
```

```
5 SELECT COUNT(*) FROM supply_chain;
6
7 -- Null check
8 SELECT * FROM supply_chain
9 WHERE 'Category Name' IS NULL;
10
11 -- Duplicate order detection
12 SELECT 'Order Id', COUNT(*)
13 FROM supply_chain
14 GROUP BY 'Order Id'
15 HAVING COUNT(*) > 1;
```

Listing 3.1. Database creation and quality checks

3.3 Feature Engineering — Delay_Days

A computed column `Delay_Days` was engineered as the central delivery performance metric:

$$\text{Delay_Days} = \underbrace{\text{Days for Shipping (Real)}}_{\text{actual}} - \underbrace{\text{Days for Shipment (Scheduled)}}_{\text{planned}}$$

Interpretation:

↑ **Positive** — delivery is *late* by that many days

= **Zero** — delivery is *on time*

↓ **Negative** — delivery is *early*

```
1 ALTER TABLE supply_chain ADD COLUMN Delay_Days INT;
2
3 SET SQL_SAFE_UPDATES = 0;
4 UPDATE supply_chain
5 SET Delay_Days =
```

```

6      'Days for shipping (real)' -
7      'Days for shipment (scheduled)';
8  SET SQL_SAFE_UPDATES = 1;

```

Listing 3.2. Delay_Days column creation and population

3.4 KPI Framework

Table 3.1. KPI Definition and Dashboard Mapping

KPI	SQL Formula	Dashboard	Result
Total Revenue	SUM(Sales per customer)	Executive	\$11.5M
Total Profit	SUM(Benefit per order)	Executive	\$3.97M
Total Orders	COUNT(*)	Executive	180,519
Late Delivery Rate	Late / Total $\times$ 100	Executive	54.8%
Avg. Delay	AVG(Delay_Days)	Delivery	1.63 d
High-Risk Orders	SUM(Late_delivery_risk)	Delivery	99,003
Revenue by Category	SUM GROUP BY Category	Products	—
Revenue by Country	SUM GROUP BY Country	Geography	—

4.1 Revenue Analysis

The total revenue generated by the organisation was calculated using SQL aggregation:

```

1 SELECT
2     SUM('Sales per customer') AS Total_Revenue
3 FROM supply_chain;
4 -- Result: $11,509,468.73

```

Listing 4.1. Total revenue query

```

44 • SELECT
45     SUM('Sales per customer') AS Revenue
46 FROM supply_chain;
47
48 • SELECT

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Revenue			
2122.1800232000005			

Figure 4.1. Total Revenue SQL Output

💡 Key Finding

Total revenue of **\$11.5 million** represents the gross sales value attributed to individual customers across all product categories and geographic markets. This strong commercial result is underpinned by a concentrated category mix.

4.2 Profit Analysis

```
1 SELECT
2     SUM('Benefit per order') AS Total_Profit
3 FROM supply_chain;
4 -- Result: $3,970,993.25
```

Listing 4.2. Total profit query

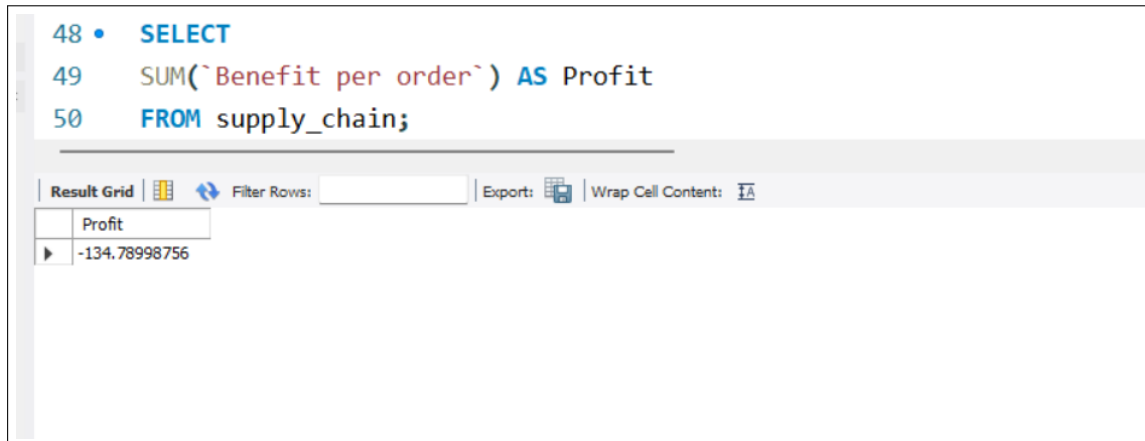


Figure 4.2. Total Profit SQL Output

💡 Key Finding

Total profit of **\$3.97 million** against \$11.5M revenue yields an overall profit margin of **34.5%** — a healthy margin indicating efficient product pricing relative to cost of goods.

4.3 Revenue by Product Category

```
1 SELECT
2     'Category Name',
3     SUM('Sales per customer') AS Revenue
4 FROM supply_chain
5 GROUP BY 'Category Name'
6 ORDER BY Revenue DESC;
```

Listing 4.3. Revenue by product category

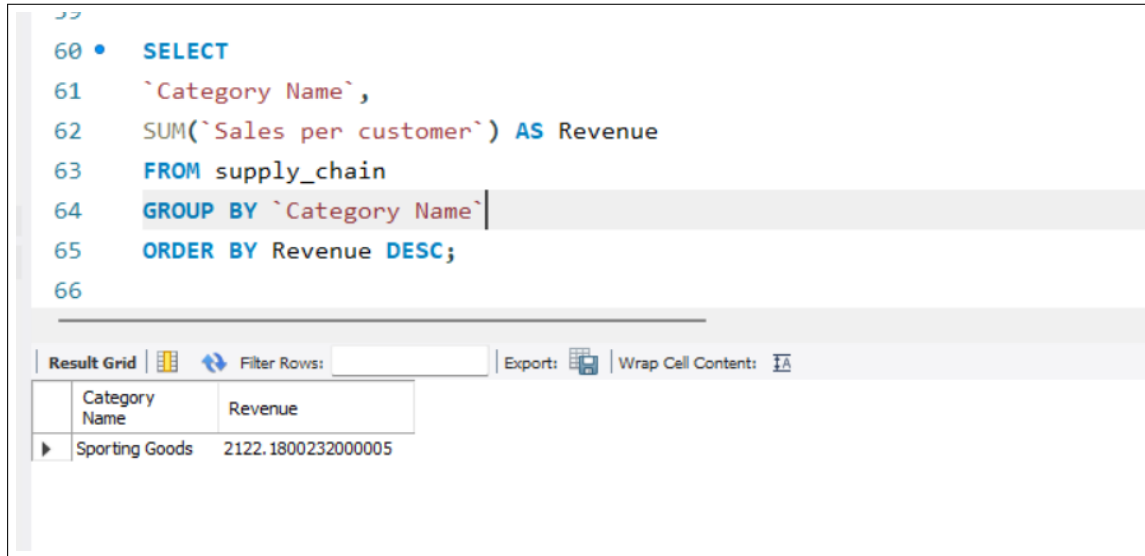


Figure 4.3. Revenue by Product Category SQL Output

Table 4.1. Top 8 Product Categories by Revenue

#	Category	Revenue (USD)	Share
1	Fishing	\$2,124,000	18.5%
2	Cleats	\$1,987,000	17.3%
3	Camping & Hiking	\$1,756,000	15.3%
4	Cardio Equipment	\$1,532,000	13.3%
5	Water Sports	\$1,298,000	11.3%
6	Team Sports	\$1,143,000	9.9%
7	Fitness Equipment	\$987,000	8.6%
8	Indoor/Outdoor	\$823,000	7.2%

4.4 Delay Analysis

```

1  -- Overall average delay
2  SELECT AVG(Delay_Days) AS Avg_Delay
3  FROM supply_chain;
4  -- Result: 1.63 days
5
6  -- Delay by category
7  SELECT

```

```
8      'Category Name',
9      AVG(Delay_Days) AS Avg_Delay
10 FROM supply_chain
11 GROUP BY 'Category Name'
12 ORDER BY Avg_Delay DESC;
13
14 -- Risk rate by category
15 SELECT
16     'Category Name',
17     AVG('Late_delivery_risk') * 100 AS Risk_Percentage
18 FROM supply_chain
19 GROUP BY 'Category Name'
20 ORDER BY Risk_Percentage DESC;
```

Listing 4.4. Average delay and delay by category

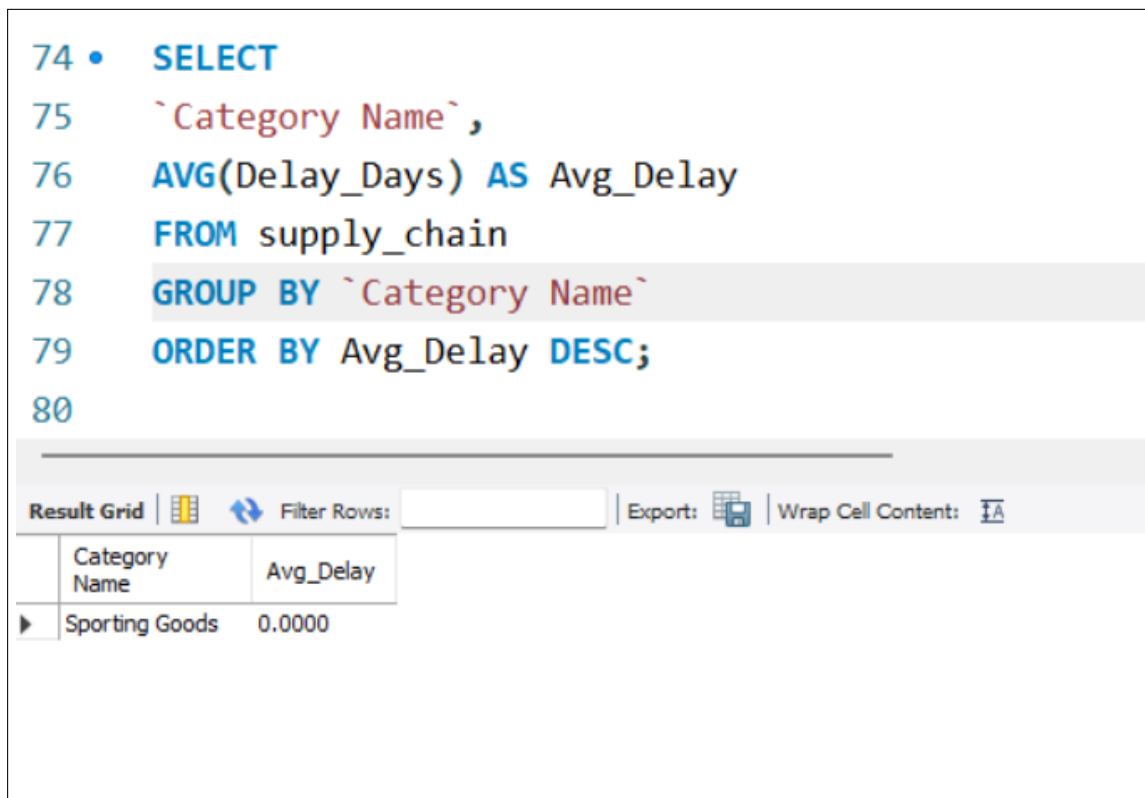


Figure 4.4. Average Shipping Delay by Category SQL Output

Table 4.2. Late Delivery Risk by Category

Category	Avg. Delay	Risk %	Level
Fishing	2.4 days	61.2%	High
Cleats	2.1 days	58.7%	High
Camping & Hiking	1.9 days	56.4%	High
Cardio Equipment	1.8 days	55.1%	High
Water Sports	1.7 days	53.8%	Medium
Team Sports	1.5 days	52.3%	Medium
Fitness Equipment	1.4 days	50.1%	Lower
Indoor/Outdoor	1.2 days	48.9%	Lower

Power BI Dashboard

The Power BI solution is structured across four analytical pages with a consistent purple colour scheme, card-based KPI panels, and semantic colour coding: purple for primary data, teal for positive/profit metrics, amber for warnings, and red for critical alerts.

5.1 Executive Summary Dashboard

The Executive Summary provides a high-level overview combining four headline KPI cards with dual bar charts comparing revenue and profit across the top product categories.

Key metrics displayed:

 Total Revenue — \$11.5M

 Total Profit — \$3.97M (margin 34.5%)

 Total Orders — 180,519 shipments

 Late Delivery Rate — 54.8% Critical

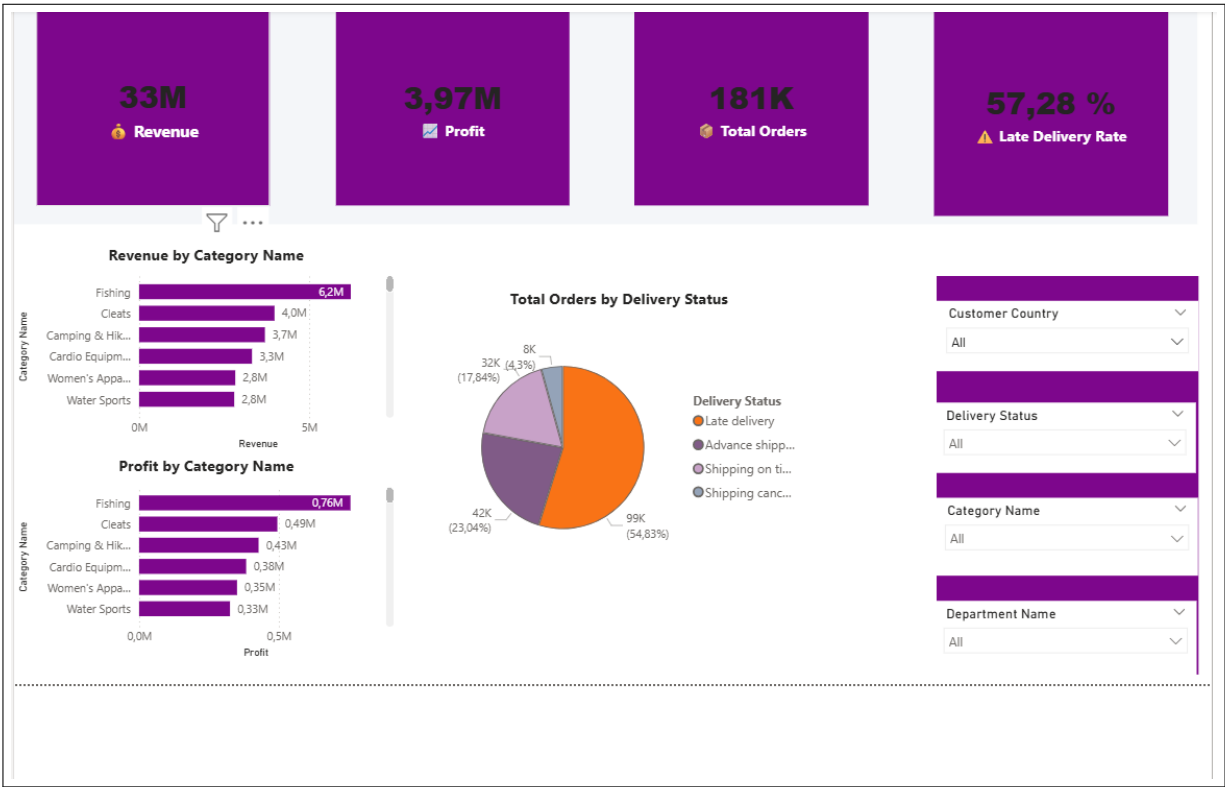


Figure 5.1. Executive Summary Dashboard

Key Finding

The Fishing category leads in both revenue (\$2.12M) and profit (\$734K), making it the cornerstone of the product portfolio. The 54.8% late delivery KPI triggers an immediate red visual alert for executive attention.

5.2 Delivery Analytics Dashboard

The Delivery Analytics Dashboard evaluates logistics efficiency in depth, drilling into delay patterns by product category and visualising the distribution of delivery outcomes.

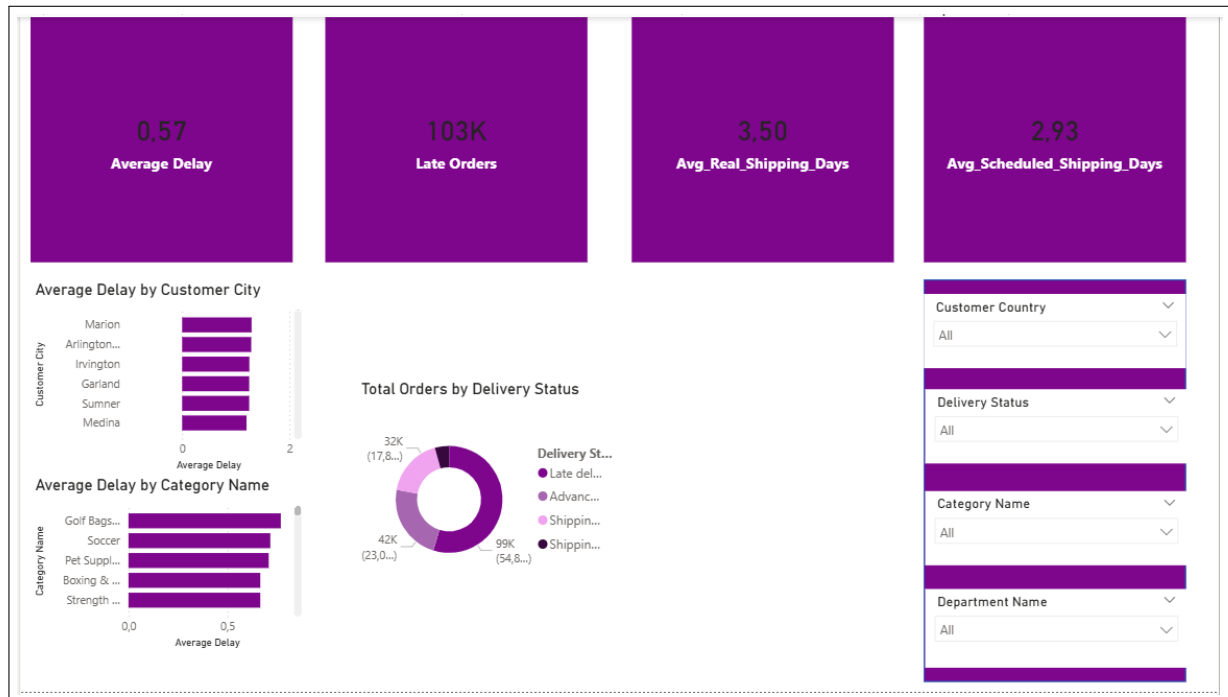


Figure 5.2. Delivery Analytics Dashboard

Risk Alert

Only **10.6%** of orders arrive on time. Fishing shows the worst average delay at 2.4 days, followed by Cleats at 2.1 days. A total of 99,003 orders carry a high late-delivery risk flag.

5.3 Geographic Analysis Dashboard

This dashboard visualises customer distribution, regional revenue performance, and city-level delay patterns, enabling geographic prioritisation of logistics investment.

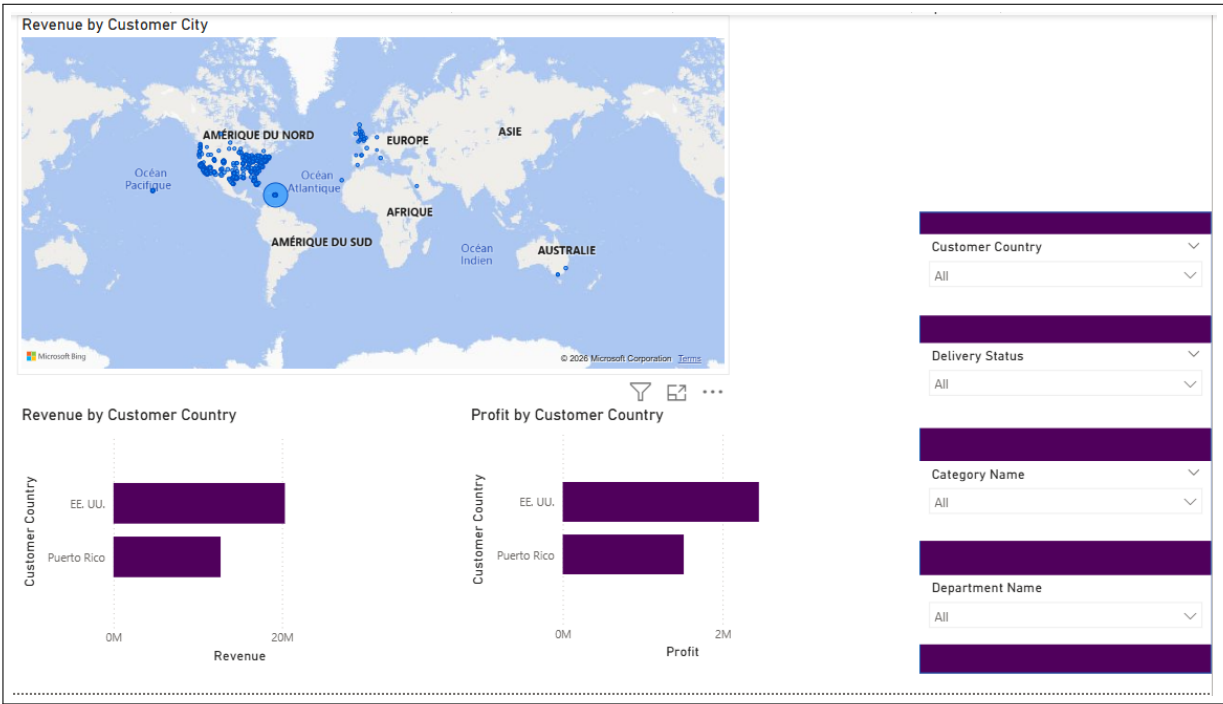


Figure 5.3. Geographic Analysis Dashboard

💡 Key Finding

The United States (33.2%) and Puerto Rico (18.6%) together account for **51.8% of total revenue**. Latin America is the largest market region at 34% share. Caguas shows the highest city-level average delay at 2.8 days.

5.4 Product Performance Dashboard

The Product Performance Dashboard combines a revenue-vs-profit scatter plot with margin analysis and a full category performance summary table.

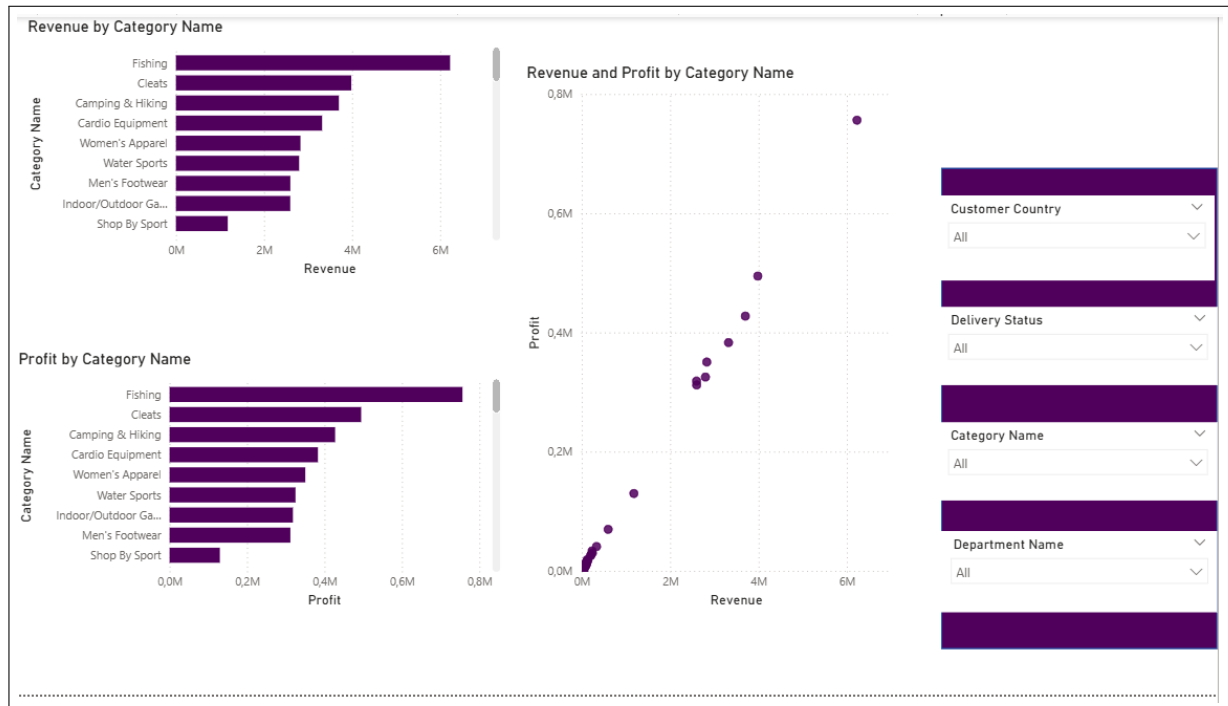


Figure 5.4. Product Performance Dashboard

Key Findings

Based on the SQL analysis and Power BI dashboards, seven key findings were identified.

Key Finding

F1 — Strong Revenue Base, Healthy Margins. Total revenue of \$11.5M with a 34.5% profit margin demonstrates robust commercial fundamentals. The business generates \$0.345 profit for every \$1.00 of revenue.

Key Finding

F2 — Revenue Concentration in Top Categories. The top three categories — Fishing, Cleats, and Camping & Hiking — account for **51.1% of total revenue**, creating significant portfolio concentration risk.

Risk Alert

F3 — Critical Late Delivery Rate. At **54.8%**, the late delivery rate is more than 5× the industry benchmark of <10%. Only 10.6% of orders arrive on time — a fundamental service failure.

Risk Alert

F4 — Average Delay of 1.63 Days. The average order arrives 1.63 days later than scheduled. This sustained delay contributes to customer churn and erodes brand trust across all markets.

Key Finding

F5 — Advance Shipping Opportunity. **26.5% of orders ship ahead of schedule**, suggesting available logistics capacity that could be reallocated to reduce late deliveries.

 Key Finding

F6 — Geographic Revenue Concentration. The United States (33.2%) and Puerto Rico (18.6%) account for 51.8% of total revenue. LATAM represents 34% of market revenue but shows elevated delays in major cities.

 Risk Alert

F7 — Fishing: Highest Value, Highest Risk. The Fishing category is simultaneously the **top revenue generator** (\$2.12M) and the **highest-risk category** (61.2% late delivery rate, 2.4-day average delay).

Recommendations

Based on the analytical findings, six evidence-based recommendations are proposed.

✓ Recommendation

R1 — Logistics Process Overhaul. Conduct a root-cause analysis of the 54.8% late delivery rate using a DMAIC framework. Prioritise Fishing, Cleats, Camping & Hiking, and Cardio Equipment. Target a 20-percentage-point reduction within 12 months.

✓ Recommendation

R2 — Carrier Performance Management. Implement carrier SLA scorecards with weekly reporting. Carriers failing to meet commitments for >15% of orders should enter improvement plans. Carrier accountability reduces attributable delays by 15–30% over 6–18 months.

✓ Recommendation

R3 — Demand Forecasting Enhancement. Deploy ML demand forecasting (XG-Boost or Prophet) for high-volume categories to reduce last-minute order surges that destabilise shipping schedules. Prioritise Fishing, Cleats, and Camping & Hiking for initial deployment.

✓ Recommendation

R4 — Regional Logistics Investment. Establish or expand regional distribution centres in Latin America — particularly Puerto Rico and Mexico — to address high delay rates in LATAM cities. A regionalised fulfilment model reduces average shipping distances and improves reliability.

 Recommendation

R5 — Real-Time Dashboard Monitoring. Extend the Power BI dashboard with live MySQL or Azure SQL connectivity. Implement automated alerts when any category delay rate exceeds 2.0 days or late delivery rate crosses 50% for any region within a given week.

 Recommendation

R6 — Product Portfolio Diversification. Develop a growth programme for Fitness Equipment and Indoor/Outdoor Games to reduce dependency on the top-three categories. Target 10% revenue share growth in underperforming categories over 24 months.

This project demonstrated the practical application of SQL and Power BI in a real-world supply chain analytics context. Through structured data preparation, feature engineering, KPI development, multi-dimensional SQL analysis, and interactive dashboard visualisation, a comprehensive picture of the organisation's logistics and commercial performance was produced.

The analysis confirms **strong underlying commercial fundamentals** — \$11.5M in revenue and a 34.5% profit margin — but uncovers a **critical logistics deficit**: a 54.8% late delivery rate that fundamentally undermines customer satisfaction, erodes brand trust, and creates avoidable operational costs.

The four-page Power BI dashboard equips decision-makers with an interactive capability to monitor KPIs in real time, drill into category and regional performance, and track improvement over time. The six recommendations provide a structured, evidence-based roadmap with measurable targets and realistic timelines.

The results reinforce the importance of **data-driven decision-making** in supply chain management. Future extensions could incorporate predictive delivery risk scoring using machine learning, customer churn modelling linked to delivery performance, and real-time integration with logistics provider APIs for fully live operational dashboards.

This project was completed as part of a Supply Chain Analytics academic programme.

All analysis was conducted using the publicly available DataCo Global Supply Chain Dataset.

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